

Microplastics results

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Microplastics

Total microplastics model

```
Family: nbinom2 ( log )
Formula:
particles ~ category + bottle_material + source_type + adv_filtration
Zero inflation:      ~.
Data: mp.total
```

AIC	BIC	logLik	-2*log(L)	df.resid
414.6	435.9	-196.3	392.6	40

Dispersion parameter for nbinom2 family (): 1.22

Conditional model:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	3.1512	0.5717	5.512	3.55e-08	***
categoryBottled	1.9115	0.6683	2.860	0.00423	**
bottle_materialPET	-1.5790	0.3624	-4.358	1.32e-05	***
bottle_materialHDPE	NA	NA	NA	NA	
source_typeSpring	-0.2986	0.4329	-0.690	0.49037	
adv_filtrationYes	0.2620	0.4951	0.529	0.59667	

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Zero-inflation model:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	1.2563	0.7636	1.645	0.0999	.
categoryBottled	-3.6909	1.5543	-2.375	0.0176	*
bottle_materialPET	1.0085	1.2452	0.810	0.4180	
bottle_materialHDPE	NA	NA	NA	NA	
source_typeSpring	-0.0125	1.1606	-0.011	0.9914	
adv_filtrationYes	-0.6730	1.1230	-0.599	0.5490	

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Confidence intervals

Negative binomial components

	2.5 %	97.5 %	Estimate
cond.(Intercept)	7.62	71.64	23.36
cond.categoryBottled	1.83	25.06	6.76
cond.bottle_materialPET	0.10	0.42	0.21
cond.source_typeSpring	0.32	1.73	0.74
cond.adv_filtrationYes	0.49	3.43	1.30

Zero-inflated components

	2.5 %	97.5 %	Estimate
zi.(Intercept)	1.27	0.06	0.28
zi.categoryBottled	843.19	1.91	40.08
zi.bottle_materialPET	4.19	0.03	0.36

zi.source_typeSpring	9.85	0.10	1.01
zi.adv_filtrationYes	17.71	0.22	1.96

Bottled model

Family: nbinom2 (log)
 Formula: particles ~ bottle_material + source_type + adv_filtration
 Zero inflation: ~.
 Data: mp.bottled.total

AIC	BIC	logLik	-2*log(L)	df.resid
369.5	384.5	-175.8	351.5	30

Dispersion parameter for nbinom2 family (): 1.19

Conditional model:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	3.3120	0.4030	8.218	< 2e-16 ***
bottle_materialHDPE	1.6313	0.3646	4.474	7.67e-06 ***
source_typeSpring	-0.1514	0.4318	-0.351	0.726
adv_filtrationYes	0.5816	0.5534	1.051	0.293

 Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Zero-inflation model:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-19.3246	7037.5151	-0.003	0.998
bottle_materialHDPE	-0.7074	1.3152	-0.538	0.591
source_typeSpring	17.7934	7037.5151	0.002	0.998
adv_filtrationYes	18.1784	7037.5152	0.003	0.998

Confidence intervals

Negative binomial components

	2.5 %	97.5 %	Estimate
cond.(Intercept)	12.45	60.46	27.44
cond.bottle_materialHDPE	2.50	10.44	5.11
cond.source_typeSpring	0.37	2.00	0.86
cond.adv_filtrationYes	0.60	5.29	1.79

Zero-inflated components

	2.5 %	97.5 %	Estimate
zi.(Intercept)	Inf	0.00	246930172.54
zi.bottle_materialHDPE	26.71	0.15	2.03
zi.source_typeSpring	Inf	0.00	0.00
zi.adv_filtrationYes	Inf	0.00	0.00

PE model

```
Family: nbinom2 ( log )
Formula:          particles ~ bottle_material + source_type + adv_filtration
Zero inflation:    ~.
Data: mp.bottled.pe
```

AIC	BIC	logLik	-2*log(L)	df.resid
325.6	340.6	-153.8	307.6	30

Dispersion parameter for nbinom2 family (): 0.897

Conditional model:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	4.0518	0.4965	8.161	3.34e-16 ***
bottle_materialPET	-1.4517	0.4692	-3.094	0.00198 **
source_typeSpring	0.5429	0.5590	0.971	0.33144
adv_filtrationYes	1.1332	0.7293	1.554	0.12026

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Zero-inflation model:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-3.8033	2.0454	-1.859	0.063 .
bottle_materialPET	2.1828	1.3356	1.634	0.102
source_typeSpring	1.3814	1.7203	0.803	0.422
adv_filtrationYes	0.3883	1.8810	0.206	0.836

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Confidence intervals

Negative binomial components

	2.5 %	97.5 %	Estimate
cond.(Intercept)	21.73	152.16	57.50
cond.bottle_materialPET	0.09	0.59	0.23
cond.source_typeSpring	0.58	5.15	1.72
cond.adv_filtrationYes	0.74	12.97	3.11

Zero-inflated components

	2.5 %	97.5 %	Estimate
zi.(Intercept)	2470.74	0.81	44.85
zi.bottle_materialPET	1.54	0.01	0.11
zi.source_typeSpring	7.32	0.01	0.25
zi.adv_filtrationYes	27.07	0.02	0.68

PET model

Family: nbinom2 (log)
Formula: particles ~ source_type + bottle_material + adv_filtration
Zero inflation: ~.
Data: mp.bottled.pet

AIC	BIC	logLik	-2*log(L)	df.resid
174.4	189.3	-78.2	156.4	30

Dispersion parameter for nbinom2 family (): 3.49

Conditional model:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	3.2052	0.4468	7.174	7.3e-13 ***
source_typeSpring	0.1953	0.4714	0.414	0.67871
bottle_materialPET	-1.0995	0.3512	-3.131	0.00174 **
adv_filtrationYes	0.4547	0.5932	0.767	0.44333

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Zero-inflation model:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	0.5609	1.0008	0.560	0.575
source_typeSpring	-0.2691	0.9643	-0.279	0.780
bottle_materialPET	0.1761	0.7583	0.232	0.816
adv_filtrationYes	-0.2334	1.1691	-0.200	0.842

Confidence intervals

Negative binomial components

	2.5 %	97.5 %	Estimate
cond.(Intercept)	10.27	59.20	24.66
cond.source_typeSpring	0.48	3.06	1.22
cond.bottle_materialPET	0.17	0.66	0.33
cond.adv_filtrationYes	0.49	5.04	1.58

Zero-inflated components

	2.5 %	97.5 %	Estimate
zi.(Intercept)	4.06	0.08	0.57
zi.source_typeSpring	8.66	0.20	1.31
zi.bottle_materialPET	3.71	0.19	0.84
zi.adv_filtrationYes	12.49	0.13	1.26

P-value adjustment

```
pvals.mp <- NULL
# Conclusion 1
# total model, category, zero inflated component
pvals.mp[1] <- 0.0176
# total model, category, conditional component
pvals.mp[2] <- 0.00423
# Conclusion 2
# Fisher exact test
pvals.mp[3] <- 1
# bottled model, source type, conditional component
pvals.mp[4] <- 0.726
# Conclusion 3
```

```

# bottled model, bottle material, zero inflated component
pvals.mp[5] <- 0.591
# bottled model, bottle material, conditional component
pvals.mp[6] <- 7.67e-06
# Conclusion 3, PE
# bottled PE model, bottle material, zero inflated component
pvals.mp[7] <- 0.102
# bottled PE model, bottle material, conditional component
pvals.mp[8] <- 0.00198
# Conclusion 3, PET
# bottled PET model, bottle material, zero inflated component
pvals.mp[9] <- 0.816
# bottled PET model, bottle material, conditional component
pvals.mp[10] <- 0.00174
# Conclusion 4
# Fisher exact test
pvals.mp[11] <- 0.5834
# bottled model, filtration, conditional component
pvals.mp[12] <- 0.293

names(pvals.mp) <- c(
  "Conclusion.mp1.1", "Conclusion.mp1.2",
  "Conclusion.mp2.1", "Conclusion.mp2.2",
  "Conclusion.mp3.1", "Conclusion.mp3.2",
  "Conclusion.mp3.3", "Conclusion.mp3.4",
  "Conclusion.mp3.5", "Conclusion.mp3.6",
  "Conclusion.mp4.1", "Conclusion.mp4.2"
)

```

Conclusion.mp1.1	Conclusion.mp1.2	Conclusion.mp2.1	Conclusion.mp2.2
0.04	0.01	1.00	0.87
Conclusion.mp3.1	Conclusion.mp3.2	Conclusion.mp3.3	Conclusion.mp3.4
0.79	0.00	0.20	0.01
Conclusion.mp3.5	Conclusion.mp3.6	Conclusion.mp4.1	Conclusion.mp4.2
0.89	0.01	0.79	0.50

Microplastics Writeup

Conclusion 1: Bottled v. municipal

Conclusion 1.1

- The odds of any microplastic particles being detected in a bottled water sample is 40.08 times higher compared to municipal water (adjusted ratio: 40.08, 95% interval for odds ratio: [1.91, 843.19]), keeping other sample characteristics fixed.
 - Unadjusted p-value: 0.0176
 - Adjusted p-value: 0.04
- With 25% of municipal samples having microplastics detected, the probability of detection for bottled water is 3.72 times higher than in municipal water (adjusted ratio: 3.72, adjusted ratio: 3.72, 95% interval for relative risk: [1.55, 3.99]).
 - Unadjusted p-value: 0.0176
 - Adjusted p-value: 0.04

Conclusion 1.2

- For samples with microplastic particles detected, the mean number of microplastic particles in a bottled water sample is 6.76 times higher compared to municipal water (adjusted ratio: 6.76, 95% interval for multiplicative effect: [1.83, 25.06]).
 - Unadjusted p-value: 0.00423
 - Adjusted p-value: 0.01
- These results should be interpreted with caution due to the small number of samples. We can see the confidence intervals are wide, denoting a large amount of uncertainty. However, overall the evidence suggests bottled water has more microplastics than municipal water on average.

Conclusion 2: Spring v. non-spring

Conclusion 2.1

- The zero-inflated portion of the model has really large standard errors, so the estimate and interval are not informative! If we were to use the zero-inflated model:
 - Among bottled water samples only, the odds of any microplastic particles being detected in a spring water sample is 100% lower compared to non-spring water (adjusted ratio: 0, p-value: 0.07, 95% interval for odds ratio: [0, ∞]), keeping other sample characteristics fixed.
 - * Unadjusted p-value: 0.998
 - With 86% of non-spring samples having microplastics detected, the probability of detection for spring water is 100% lower than in non-spring water (adjusted ratio: 0, 95% interval for relative risk: [0, NaN]).
 - * Unadjusted p-value: 0.998
- We instead consider an Exact Two-Sample Fisher-Pitman Permutation Test.

- Unadjusted p-value: 1
- Adjusted p-value: 1
- The detection rate in spring water was 84 %.
- The detection rate in non-spring water was 86 %.

Conclusion 2.2

- For bottled water samples with microplastic particles detected, the mean number of microplastic particles in a spring water sample is 14% lower compared to non-spring water, but the difference is not significant (adjusted ratio: 0.86, 95% interval for multiplicative effect: [0.37, 2]).
 - Unadjusted p-value: 0.726
 - Adjusted p-value: 0.87
- Overall, the evidence suggests there is little or no difference between spring and non-spring sources.

Conclusion 3: Total MP, bottle material

Conclusion 3.1

- The odds of any microplastic particles being detected in a HDPE bottle sample is 2.03 times higher compared to PET bottle samples (adjusted ratio: 2.03, 95% interval for odds ratio: [0.15, 26.71]), keeping other sample characteristics fixed.
 - Unadjusted p-value: 0.591
 - Adjusted p-value: 0.79
- With 81% of PET bottle samples having microplastics detected, the probability of detection for HDPE bottles is 10% higher than in PET bottles (adjusted ratio: 1.1, 95% interval for relative risk: [0.5, 1.22]).
 - Unadjusted p-value: 0.591
 - Adjusted p-value: 0.79

Conclusion 3.2

- For samples with microplastic particles detected, the mean number of microplastic particles in HDPE bottles is 5.11 times higher compared to PET bottle samples (adjusted ratio: 5.11, 95% interval for multiplicative effect: [2.5, 10.44]).
 - Unadjusted p-value: 7.67e-06
 - Adjusted p-value: 0

Conclusion 3: PE, bottle material

Conclusion 3.3

- The odds of any PE particles being detected in a PET bottle sample is 89% lower compared to HDPE bottle samples (adjusted ratio: 0.11, 95% interval for odds ratio: [0.01, 1.54]).
 - Unadjusted p-value: 0.102
 - Adjusted p-value: 0.2
 - The difference is not significant.
- With 92% of HDPE bottle samples having PE microplastics detected, the probability of PE detection for PET bottles is 40% lower than in HDPE bottles (adjusted ratio: 0.6, 95% interval for relative risk: [0.09, 1.03]).
 - Unadjusted p-value: 0.102
 - Adjusted p-value: 0.2
 - The difference is not significant.

Conclusion 3.4

- For samples with PE particles detected, the mean number of PE particles in PET bottles is 77 % lower compared to HDPE bottle samples (adjusted ratio: 0.23, 95% interval for multiplicative effect: [0.09, 0.59]).
 - Unadjusted p-value: 0.00198
 - Adjusted p-value: 0.01

Conclusion 3: PET, bottle material

Conclusion 3.5

- The odds of any PET particles being detected in a PET bottle sample is 16% lower compared to HDPE bottle samples (adjusted ratio: 0.84, 95% interval for odds ratio: [0.19, 3.71]), keeping other sample characteristics fixed, but the difference is not significant.
 - Unadjusted p-value: 0.816
 - Adjusted p-value: 0.89
 - The difference is not significant.
- With 42% of HDPE bottle samples having microplastics detected, the probability of PET detection for PET bottles is 10%% higher than in HDPE bottles (adjusted ratio: 0.9, 95% interval for relative risk: [0.29, 1.74]).
 - Unadjusted p-value: 0.816
 - Adjusted p-value: 0.89

- The difference is not significant.

Conclusion 3.6

- For samples with microplastic particles detected, the mean number of PET particles in PET bottles is 67% lower compared to HDPE bottle samples (adjusted ratio: 0.33, 95% interval for multiplicative effect: [0.17, 0.66]).
 - Unadjusted p-value: 0.00174
 - Adjusted p-value: 0.01

Summary

- Overall, there is strong evidence HDPE bottle samples have more total microplastic particles on average compared to PET bottles. There is also evidence HDPE bottles have more PE and PET particles compared to PET bottles on average.

Conclusion 4, advanced filtration

Conclusion 4.1

- The zero-inflated portion of the model has really large standard errors, so the estimate and interval are not informative! If we were to use the zero-inflated model:
 - Among bottled water samples only, the odds of any microplastic particles being detected in a sample with advanced filtration is 100% lower compared to a sample without, keeping other sample characteristics fixed, but the difference is not significant (adjusted ratio: 0, 95% interval for odds ratio: $[0, \infty]$).
 - * Unadjusted p-value: 0.998
 - With 87% of samples without advanced filtration having microplastics detected, the probability of detection for samples with advanced filtration is 100% lower than in samples without, but the difference is not significant (adjusted ratio: 0, p-value: 95% interval for relative risk: $[0, \text{NaN}]$).
 - * Unadjusted p-value: 0.998
- Instead we report the p-value from Exact Two-Sample Fisher-Pitman Permutation Test.
 - Unadjusted p-value: 0.5834
 - Adjusted p-value: 0.79
 - The detection rate with advanced filtration was 75 %.
 - The detection rate without advanced filtration was 87 %.

Conclusion 4.2

- For samples with microplastic particles detected, the mean number of microplastic particles in a sample with advanced filtration is 1.79 times higher compared to a sample without, but the difference is not significant (adjusted ratio: 1.79, 95% interval for multiplicative effect: [0.6, 5.29]).
 - Unadjusted p-value: 0.293
 - Adjusted p-value: 0.5
- Overall, the evidence suggests there is little or no difference between having advanced filtration or not.